

# Integral Square Error Evaluation

## Definition of the transfer function

```
In[9]:= H[s_] := (s^2 + K/2 * s) / (s^2 + K/2 * s + 5/2)
```

```
In[10]:= H[s]
```

```
Out[10]=
```

$$\frac{\frac{Ks}{2} + s^2}{\frac{5}{2} + \frac{Ks}{2} + s^2}$$

## Calcul of the response of the system to an unit step disturbance signal

```
In[11]:= y[t_] := InverseLaplaceTransform[H[s] / s, s, t, Assumptions -> {t >= 0, K > 0, K < 40}]
```

[transformée de Laplace inverse]

[hypothèses]

```
In[12]:= y[t]
```

```
Out[12]=
```

$$\frac{-e^{\left(-\frac{K}{4}-\frac{1}{4}\sqrt{-40+K^2}\right)t}K + e^{\left(-\frac{K}{4}+\frac{1}{4}\sqrt{-40+K^2}\right)t}K + e^{\left(-\frac{K}{4}-\frac{1}{4}\sqrt{-40+K^2}\right)t}\sqrt{-40+K^2} + e^{\left(-\frac{K}{4}+\frac{1}{4}\sqrt{-40+K^2}\right)t}\sqrt{-40+K^2}}{2\sqrt{-40+K^2}}$$

## Calcul of the ISE

```
In[15]:= ise = Integrate[y[t]^2, {t, 0, Infinity}, Assumptions -> 0 < K < 40]
```

[intégrer]

[infini]

[hypothèses]

```
Out[15]=
```

$$\frac{1}{K} + \frac{K}{10}$$

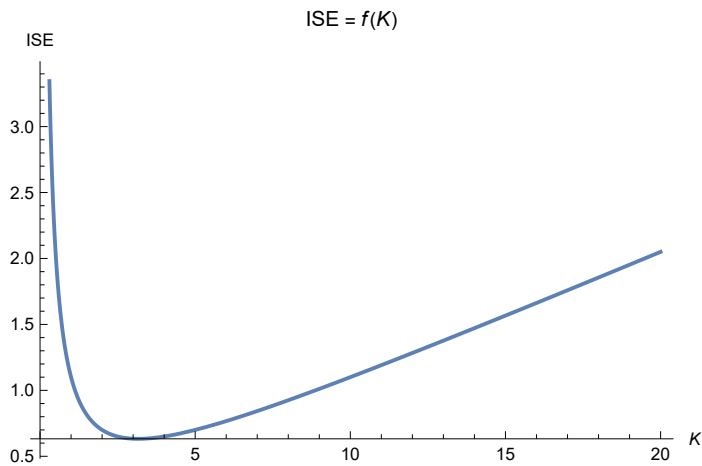
```
In[16]:= ise
```

```
Out[16]=
```

$$\frac{1}{K} + \frac{K}{10}$$

```
In[25]:= Plot[ise, {K, .1, 20}, AxesLabel → {HoldForm[K], HoldForm[ISE]},
  |tracé |étiquette des a... |maintiens forme |maintiens forme
  PlotLabel → HoldForm[ISE = f[K]], LabelStyle → {GrayLevel[0]}]
  |étiquette de t... |maintiens forme |style d'étiquette |niveau de gris
```

Out[25]=



```
In[74]:= Kstar = Assuming[40 > K > 0, FindArgMin[ise, K]]
  |supposant |trouve argument de minimum
```

Out[74]=

{ 3.16228 }

---

## Evaluation of the tracking error du to the distirbuance signal

```
In[69]:= ystar[t_] := y[t] /. {K → Kstar}
```

```
In[70]:= ystar[t]
```

Out[70]=

$$\left\{ (0. - 0.0912871 i) \left( (-3.16228 + 5.47723 i) e^{(-0.790569 - 1.36931 i) t} + (3.16228 + 5.47723 i) e^{(-0.790569 + 1.36931 i) t} \right) \right\}$$

```

In[71]:= Plot[-ystar[t], {t, 0, 20}, PlotRange -> {{0, 20}, {-1, 1}},
  tracé zone de tracé
  AxesLabel -> {HoldForm[t], HoldForm[ $\epsilon[t]$ ]},
  étiquette des a... maintiens forme maintiens forme
  PlotLabel -> HoldForm[ $\epsilon[t]$ ], LabelStyle -> {GrayLevel[0]}
  étiquette de t... maintiens forme style d'étiquette niveau de gris

```

Out[71]=

